

## Initial distance to *Kalmia angustifolia* as a predictor of planted conifer growth

Nelson Thiffault · Geneviève Picher · Isabelle Auger

Received: 28 October 2011 / Accepted: 3 April 2012 / Published online: 22 April 2012  
© Her Majesty the Queen in Right of Canada 2012

**Abstract** In the Canadian boreal forest, conifer plantations are mainly used to overcome poor natural regeneration following harvesting or wildfires. However, competitive interactions can impair the successful establishment of forest plantations, especially in the presence of ericaceous species, such as *Kalmia angustifolia*, that are good competitors for soil resources. We used data from a silvicultural trial established in northwestern Quebec (Canada) to test the hypothesis that *Kalmia*'s effect on seedling growth is an asymptotic nonlinear function of proximity to *Kalmia* at time of planting. Our main objective was to derive species-specific thresholds of *Kalmia*-proximity tolerance at the time of planting for boreal sites prone to *Kalmia* invasion following harvesting. Using nonlinear quantile regressions (90th percentile), height, diameter and growth of planted *Picea mariana* and *Pinus banksiana* seedlings over 8 years was regressed against the initial distance to the nearest *Kalmia* stem at time of planting. We also investigated the relation between annual growth and the distance to nearest *Kalmia* stem during any given year between age 2 and 8-years. Finally, we evaluated how mechanical soil scarification influenced *Kalmia* recovery over time by looking at its rate of spread, estimated from mean distance to planted seedlings over 8 years. Silvicultural treatments increased the intercepts and asymptotes of all variables for *Pinus*. However, asymptotic values for *Picea* could not be identified, thus precluding threshold assessment for initial distance to *Kalmia* at planting for this species. For *Pinus* planted on scarified sites, a *Kalmia*-free radius of 60 cm would lead to ~80 to 85 % of the 90th percentile of maximum potential height at age 8 years. Mechanical scarification created *Kalmia*-free microsites that were rapidly re-invaded by spread of *Kalmia*; planting immediately following soil preparation of these sites is therefore important for ensuring successful plantation establishment.

---

N. Thiffault · G. Picher · I. Auger  
Direction de la recherche forestière, Ministère des Ressources naturelles et de la Faune du Québec,  
2700 rue Einstein, Québec, QC G1P 3W8, Canada

N. Thiffault (✉)  
Centre d'Étude de la Forêt, Faculté de foresterie, de géographie et de géomatique,  
Université Laval, Québec, QC G1V 0A6, Canada  
e-mail: nelson.thiffault@mrnf.gouv.qc.ca

**Keywords** Reforestation · Scarification · Boreal forest · Ericaceous shrubs · Quantile regression

## Introduction

Forest plantations are an important component of sustainable forest management (Evans 2009). They can be used to meet a wide variety of social, economic and environmental objectives and to provide key ecosystem services (Paquette and Messier 2010), such as increasing species diversity and cover (Ganatsas et al. 2012). In the Canadian boreal forest, conifer plantations are most often used when natural regeneration following harvesting or wildfires is inadequate and hence prescribed regeneration standards are not met (Thompson and Pitt 2003). As a result, ~605 M seedlings are planted each year across Canada, with 75 % planted in Quebec, Ontario and British Columbia (Canadian Council of Forest Ministers 2009).

Reforestation success largely depends on the management of competitive interactions between the planted species and surrounding vegetation (Newton 2006). Although facilitation interactions can be observed and promoted in some situations, disturbances usually favour the establishment and growth of opportunistic, fast-growing species that capture resources at the expense of planted trees (Balandier et al. 2006). In the boreal forest of northeastern Canada, ericaceous species compete with conifers on many regenerating sites. *Kalmia angustifolia* L. is a shade-intolerant, low understory shrub that primarily reproduces through vegetative growth from buried rhizomes and is hence well adapted to site conditions caused by wildfires and harvesting activities (Mallik 1993). *Kalmia* affects regenerating conifers by altering humus quality (Joanisse et al. 2009), by direct competition for nutrients (Thiffault et al. 2004b), and through potential allelopathic effects (Yamasaki et al. 2002; Zhu and Mallik 1994). Hence, conifers outplanted on *Kalmia*-invaded sites often exhibit symptoms of growth check and chlorotic foliage (de Montigny and Weetman 1990; Yamasaki et al. 1998).

As observed with other ericaceous species such as *Rhododendron flavum* Don. (Yildiz et al. 2010), *Kalmia* can be controlled effectively after application of chemical herbicides (Moroni et al. 2009). However, mechanical scarification is also an appropriate alternative to chemical control for promoting seedling growth on *Kalmia*-dominated sites (Thiffault et al. 2010); it is thus the preferred silvicultural tool for management of these ecosystems in Quebec, where the use of herbicide is banned on public forest lands (~90 % of the provincial forest land base in Quebec; Thiffault and Roy 2011).

Trench scarification equipment creates ditches of exposed mineral soil, flanked by a berm of mixed organic–mineral material (Prévost 1992). Specific planting locations (microsite planting) are chosen to ensure the best local conditions for the planted conifers; on mesic sites, tree planters are thus directed to position seedlings at the trench-berm interface (Burton et al. 2000; Heineman 1998). Although both the humus layer and *Kalmia* plants are severely disrupted by scarification, typical treatments leave intact bands of forest floor and vegetation about 2–3 m wide between trenches (Sutherland and Foreman 1995). Planted black spruce (*Picea mariana* (Mill.) BSP) seedlings growing within one meter of *Kalmia* plants have reduced foliar P concentration, are more often infected by pseudo-mycorrhiza and exhibit reduced growth, compared to seedling growing farther away (Yamasaki et al. 1998). Thus, recent concerns have arisen concerning the optimum planting position on scarified sites relative to *Kalmia* abundance. Moreover, long-term

results from field trials suggest that seedling sensitivity to the presence of *Kalmia* is species-specific; species with high potential rates of resource capture, such as jack pine (*Pinus banksiana* Lamb.), would be less affected by *Kalmia* as compared to species adapted to low-resource environments, such as black spruce (Thiffault et al. 2004a; Thiffault et al. 2010). Thus, the best planting microsite relative to *Kalmia* distribution is expected to vary with the planted species, and general guidelines must be adapted accordingly to favour optimum seedling growth.

We thus hypothesized that the effect of *Kalmia* on planted jack pine and black spruce growth over 8 years and dimensions at age 8-years is a monotonically increasing, nonlinear asymptotic function of the distance to the nearest *Kalmia* plant at the time of planting. This hypothesis was inspired by the work of Yamasaki (1999), who tested a nonlinear asymptotic function using the 7th-year growth increment of black spruce seedlings and distance to *Kalmia* during that same year. However, this relationship has yet to be evaluated to derive species-specific thresholds of *Kalmia*-proximity tolerance at the time of planting. Our objective was thus to test the initial (time of planting) distance to *Kalmia* as a predictor of seedling dimensions and growth over 8 years, and potentially provide foresters with an additional criteria for microsite selection.

To do so, we analyzed data from an experimental plantation established in boreal Quebec using nonlinear quantile regressions. We focussed our analytical efforts on the seedlings that were most likely to become dominant or codominant in the future stands. Annual growth data were also regressed against the corresponding distances to *Kalmia*, to better understand the distance effect over time. We also evaluated how mechanical site preparation influenced *Kalmia* recovery over time by looking at the evolution of the distance between *Kalmia* and planted seedlings over 8 growing seasons. We expected black spruce to be more sensitive to *Kalmia* proximity than jack pine. We also posited that an increase in soil perturbation intensity (double-pass scarification at 90° to each other) would be more effective in sustaining the presence of *Kalmia*-free microsites over time relative to single-pass scarification.

## Materials and methods

### Site description

The study site was located in the Abitibi region of northwestern Quebec (48°29'37"N, 76°55'40"W) within the balsam fir (*Abies balsamea* (L.) Mill.)–white birch (*Betula papyrifera* Marsh.) bioclimatic domain described by Saucier et al. (2009). The regional climate is subpolar subhumid continental, with a mean annual temperature of 2.6 °C, a growing season of 150–160 days, and a mean annual precipitation of 950 mm, of which 30 % falls as snow (Robitaille and Saucier 1998). The previous stand was 70-year-old and composed of black spruce and jack pine. It was clearcut-harvested during the summer 1999. The soil is a humo-ferric podzol (Soil Classification Working Group 1998), and characterized by an 8-cm deep mor humus. It developed on a moderately well-drained fluvioglacial deposit with a loamy-sand texture (74 % sand, 22 % silt, 4 % clay) in surface mineral horizons. The corresponding ecotype is RE21, based on the provincial classification system (Saucier et al. 2009). Ground vegetation at time of planting (June 2000; see below) was dominated by *Kalmia*, *Vaccinium angustifolium* Ait., *V. myrtilloides* Michx. and ground lichens (*Cladina* spp.) (Thiffault et al. 2005).

## Experimental design and treatments

We established an experimental plantation as a complete block split-plot field trial, with treatments randomly applied within 10 replicate blocks (see Thiffault et al. 2005 for details). In summary, we applied one of the 3 following scarification treatments in October 1999 within each block ( $30 \times 66$  m), to each of 3 main plots ( $18 \times 30$  m and separated by 6-m buffers): (a) no scarification (S0); (b) single-pass scarification with a TTS passive disk trencher (S1); and (c) double-pass scarification, with the second pass perpendicular to the first (S2). We planted 72 black spruce seedlings and 72 jack pine seedlings in the hinge position (trench-berm interface) in each main plot in early June 2000. In control plots (no trench), seedling plugs were completely inserted in the mineral soil (the root-collar located at the mineral/organic interface). Seedlings were grown over 2 years in containers of 45 cells of  $110 \text{ cm}^3$  each. For each species, we planted seedlings 1-m apart in 3 rows of 24 seedlings each, with 2 m between rows. Main plots were split longitudinally into 2 subplots; one subplot was chosen at random as a control (F0) and the other for fertilization (F1) at the time of planting with a slow-release fertilizer package (26–12–6 Silva Pak; Reforestation Technologies International, Salinas, CA, USA). Silva Paks were buried 5-cm deep, about 2-cm away from the seedling root plug. One seedling out of 2 was tagged (alternate seedlings) for long-term growth measurements (i.e. 18 seedlings per experimental unit). We used untagged seedlings for destructive sampling during the first 3 years (Thiffault et al. 2005); the resulting plantation was thus spaced at  $2 \times 2$  m.

## Seedling measurements

We measured seedling height (cm) and ground-level diameter (GLD; mm) immediately after planting (June 2000) and 1, 2, 3, 5, 6 and 8 growing seasons following planting (October 2000, 2001, 2002, 2004, 2005 and 2007). In June 2000, black spruce seedlings were  $28.5 \pm 4.6$  cm tall (mean  $\pm$  SD) and  $3.2 \pm 0.5$  mm in GLD; jack pine seedlings were  $31.5 \pm 5.6$  cm tall, and  $4.3 \pm 0.6$  mm in GLD (Thiffault et al. 2005). Absolute growth rate in height ( $\text{AGR}_H$ ) and GLD ( $\text{AGR}_D$ ) over 8 growing seasons (June 2000 to Oct. 2007) was calculated as  $(\text{height}_{\text{October 2007}} \text{ or } \text{GLD}_{\text{October 2007}} - \text{height}_{\text{June 2000}} \text{ or } \text{GLD}_{\text{June 2000}})/8$ . Annual absolute growth rate in height ( $\text{AGR}_{H\text{-annual}}$ ) was calculated as well, using data from consecutive measurements. The distance between each tagged seedling and the nearest *Kalmia* stem ( $> 0.1$  m in height; within a 3-m radius) was measured at the time of planting (June 2000) and 1, 2, 3, 5, 6, and 8 growing seasons following planting.

## Model parameterization and statistical analyses

Based on Yamasaki (1999), we assumed that (a) seedlings' height, diameter and growth are not expected to increase infinitely with increasing initial distance to *Kalmia*, and (b) seedlings in close proximity to *Kalmia* are expected to present positive values of height, GLD and growth (intercept  $> 0$ ). Using the NLIN procedure of SAS 9.2 (SAS Institute Inc. 2010), we thus performed preliminary analyses and tested functions with horizontal asymptote and non-zero ordinate-intercept (Pienarr and Turnbull 1973; Richards 1959; Schumacher 1939; Yamasaki 1999). All analyses were performed separately for jack pine and black spruce. Based on early results obtained on the same site (Thiffault et al. 2005), data from S1 and S2 plots were pooled for all the analyses of seedling dimensions and growth. Values of the model coefficients were related to the scarification  $\times$  fertilization

interaction when needed. Model selection was based on the goodness of fit with the following criteria: root mean square error (RMSE), coefficient of determination ( $R^2$ ), and on graphical evaluations. With these criteria, Yamasaki's equation (Eq. 1) was retained:

$$Y_i = a + a_1 I_1 + a_2 I_2 + a_3 I_3 + b(1 - c^{\text{DIST}_i}) \quad (1)$$

where,  $Y_i$ : Height in 2007, GLD in 2007,  $\text{AGR}_H$  or  $\text{AGR}_D$  over 8 years, for the  $i$ th seedling,  $I_1 : \begin{cases} 1, & \text{if S0F0 plots} \\ 0, & \text{otherwise} \end{cases}$ ,  $I_2 : \begin{cases} 1, & \text{if S0F1 plots} \\ 0, & \text{Otherwise} \end{cases}$ ,  $I_3 : \begin{cases} 1, & \text{if S1F0 or S2F0 plots} \\ 0, & \text{Otherwise} \end{cases}$ ,  $a$ ,  $a_1$ ,  $a_2$ ,  $a_3$ : coefficients related to the intercept,  $b$ : coefficient associated to the length from the intercept to the horizontal asymptote,  $c$ : coefficient related to the amount of curvature in the fitted curve,  $\text{DIST}_i$ : initial distance from the  $i$ th seedling to the nearest *Kalmia* stem >0.1 m in height.

Height, GLD,  $\text{AGR}_H$  and  $\text{AGR}_D$  over 8 years are expected to be influenced by an array of factors, such as soil temperature, soil water availability, and mineralization rates (Margolis and Brand 1990). As growth and yield models for plantations in Quebec are based on dominant and codominant trees (Prégent and Végiard 2000), we focussed on determining the influence of the initial distance to *Kalmia* on the 90th percentile expected responses of the planted seedlings (the seedlings that will likely become dominant or codominant). Focusing on the upper zone of the response variables plotted as functions of one independent variable, such as the initial distance to *Kalmia*, enables to identify the maximum response when other variables are not limiting (Jarvis 1976). As an example, Grossnickle and Arnott (1992) used boundary line analyses to model western hemlock (*Tsuga heterophylla* (Raf.) Sarg.) maximum photosynthesis and stomatal conductance as a function of photosynthetically active radiation or vapor pressure deficit. Similarly, quantile regression provides the opportunity for estimating models for conditional quantile functions (Koenker 2005), without the need to arbitrarily segment samples into subsets defined according to the conditioning covariates (Koenker and Hallock 2001). Like ordinary regression, all observations in quantile regression participate equally in determining the coefficients (Koenker 2005), except that conditional quantile is modelled instead of conditional mean. Thus, we estimated the superior decile of height, GLD,  $\text{AGR}_H$  and  $\text{AGR}_D$  over 8 growing seasons (2000–2007) for jack pine and black spruce using nonlinear quantile regressions based on Eq. 1, with the *nlrq* function of the *quantreg* package available in R (Koenker 2009; R Core Team 2011). When an horizontal asymptote was not reached (i.e. parameter  $b$  of Eq. 1 not significant), linear quantile regression was used with the following equation:

$$Y_i = a + a_1 I_1 + a_2 I_2 + a_3 I_2 + b \text{DIST}_i \quad (2)$$

where parameters are as in Eq. 1, except for  $b$ , which is related to the slope.

We used a linear mixed model (MIXED procedure of SAS; Littell et al. 2006) to model the annual AGR in height ( $\text{AGR}_{H\text{-annual}}$ ) for jack pine and black spruce as a function of the distance to nearest *Kalmia* stem during the corresponding growing season. Model coefficients (intercept and slope) were related to the scarification treatments (S0, S1–S2), fertilization treatments (F0, F1), time since treatment application (growing seasons 2, 3, 4–5, 6 and 7–8), and their interactions. We based the model selection on a backward method: starting with the full model and eliminating non significant effects ( $p > 0.05$ ) one at a time, beginning with the less significant. Random effects (block, block  $\times$  scarification and block  $\times$  scarification  $\times$  fertilization) are normally distributed and uncorrelated. We took

into account the covariance pattern of the repeated measurements on the same tree by specifying the structure of the residual matrix (Brown and Prescott 2006). Residual plots were used to check normality of the random effects and to identify potential outliers. A simulation-based approach was used to test the effect of the distance to the nearest *Kalmia* ( $\alpha = 0.05$ ).

The planted species and fertilization treatments were not expected to influence *Kalmia* dynamics (Moroni et al. 2009; Thiffault et al. 2010). A generalized linear mixed model (GLIMMIX procedure of SAS) was thus used to model the distance from the nearest *Kalmia* stem as a function of the scarification treatments (S0, S1, S2) and time since treatment application. This approach was selected to take into account the lack of normality of the response variable and the hierarchical structure of the data with the inclusion of random effects in the model. We used a gamma distribution with a *log* link, as suggested for positive continuous variables that are highly skewed to the right (SAS Institute Inc. 2007). In this type of model, random effects (block and block  $\times$  scarification) are normally distributed and uncorrelated (Brown and Prescott 2006). The covariance pattern of the repeated measurements was taken into account by specifying the structure of the residual matrix (Brown and Prescott 2006). Residual plots were thus used to check the normality of the random effects and to identify potential outliers. A simulation-based approach was used to separate treatment means when ANOVAs showed significant effects ( $p < 0.05$ ).

## Results

### Jack pine

We were able to model jack pine height, GLD,  $AGR_H$  and  $AGR_D$  using nonlinear quantile regressions (Table 1). Fertilization alone at the time of planting increased the model intercept by 29, 38, 31 and 47 % for jack pine height, GLD,  $AGR_H$  and  $AGR_D$ , respectively, compared to the control (unscarified and unfertilized) conditions (Figs. 1, 2). At the closest distance to *Kalmia*, the model predicted 36–57 % increases in jack pine 90th percentile height, GLD,  $AGR_H$  and  $AGR_D$  when planted in scarified (unfertilized) plots compared to unscarified plots. The combine effect of fertilization and scarification led to a 55 % increase in minimal (intercept value) 90th percentile jack pine height, and increased the expected minimal (intercept value) 90th percentile GLD by 71 %, compared to control conditions (Figs. 1, 2). When used alone, fertilization and scarification had similar impacts on the maximum predicted 90th percentile height, GLD,  $AGR_H$  and  $AGR_D$  (asymptotes), with increases ranging from 15 to 22 %, compared to the absence of treatment. The maximum predicted height increased by 28 % (83 cm) in scarified—fertilized plots, compared to control conditions (Fig. 1).

Scarification, fertilization and time since treatment application influenced the effect that distance to the nearest *Kalmia* stem had on jack pine yearly growth (significant interactions in Table 2). In non-fertilized plots (with or without scarification),  $AGR_{H-annual}$  was linearly related to the distance of the nearest *Kalmia* stem during growing seasons 2, 3 and 4–5 (Table 3, Fig. 3). The  $AGR_{H-annual}$  was linearly related to the distance of the nearest *Kalmia* stem during growing seasons 3 and 4–5 in fertilized—non-scarified plots. In scarified—fertilized plots, no significant relation was detected between the distance to nearest *Kalmia* stem and  $AGR_{H-annual}$ .

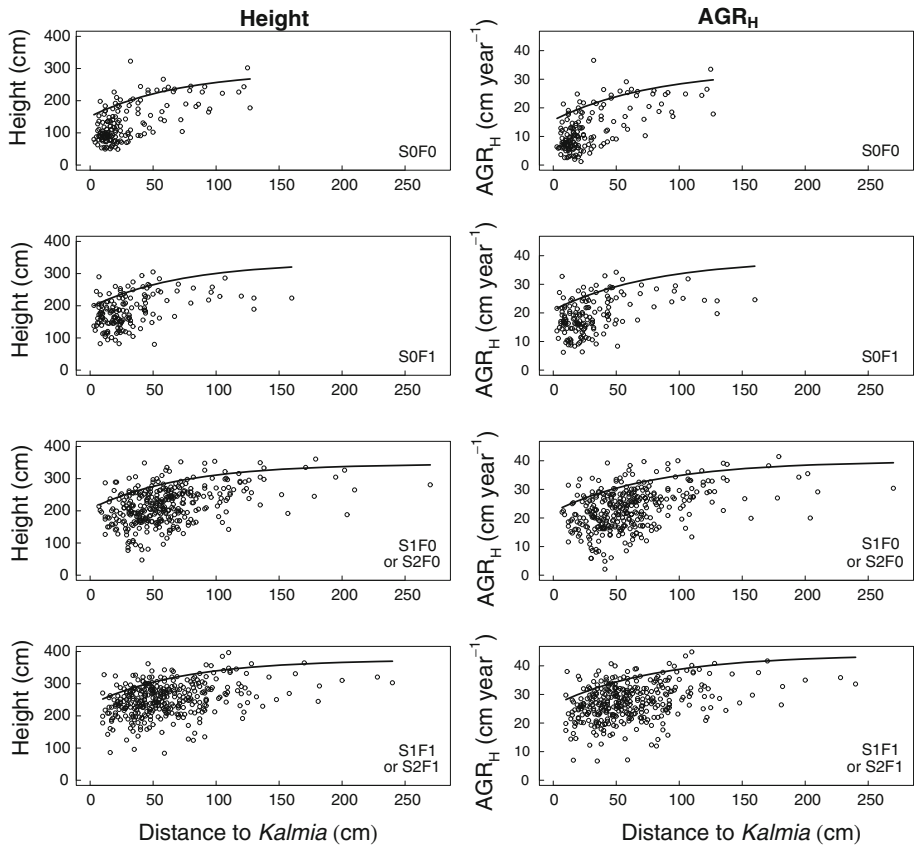
**Table 1** Parameter estimates and standard errors (SE) for *Pinus banksiana* and *Picea mariana* maximum 8th-year height, ground-level diameter (GLD), absolute growth rate in height over 8 years (AGR<sub>H</sub>) and absolute growth rate in ground-level diameter over 8 years (AGR<sub>D</sub>) on a clearcut site in northwestern Quebec (Canada)

| Variable         | Parameter             | <i>Pinus banksiana</i> |       | <i>Picea mariana</i> |       |
|------------------|-----------------------|------------------------|-------|----------------------|-------|
|                  |                       | Estimate               | SE    | Estimate             | SE    |
| Height           | <i>a</i>              | 234.02                 | 12.02 | 138.33               | 6.43  |
|                  | <i>a</i> <sub>1</sub> | −83.36                 | 6.18  | −67.75               | 7.58  |
|                  | <i>a</i> <sub>2</sub> | −39.41                 | 7.06  | −58.50               | 6.01  |
|                  | <i>a</i> <sub>3</sub> | −28.83                 | 5.15  | −36.83               | 7.74  |
|                  | <i>b</i>              | 140.84                 | 15.84 | 0.708                | 0.088 |
|                  | <i>c</i>              | 0.986                  | 0.004 | –                    | –     |
| GLD              | <i>a</i>              | 41.40                  | 3.05  | 26.72                | 1.18  |
|                  | <i>a</i> <sub>1</sub> | −17.12                 | 2.37  | −14.41               | 1.85  |
|                  | <i>a</i> <sub>2</sub> | −8.04                  | 1.82  | −9.22                | 1.52  |
|                  | <i>a</i> <sub>3</sub> | −5.18                  | 1.42  | −6.18                | 1.35  |
|                  | <i>b</i>              | 35.98                  | 3.32  | 0.178                | 0.013 |
|                  | <i>c</i>              | 0.982                  | 0.004 | –                    | –     |
| AGR <sub>H</sub> | <i>a</i>              | 26.10                  | 1.52  | 13.34                | 0.69  |
|                  | <i>a</i> <sub>1</sub> | −10.45                 | 0.93  | −8.57                | 0.93  |
|                  | <i>a</i> <sub>2</sub> | −5.13                  | 1.03  | −7.01                | 0.86  |
|                  | <i>a</i> <sub>3</sub> | −3.95                  | 0.76  | −4.14                | 0.83  |
|                  | <i>b</i>              | 17.76                  | 3.02  | 0.092                | 0.011 |
|                  | <i>c</i>              | 0.988                  | 0.005 | –                    | –     |
| AGR <sub>D</sub> | <i>a</i>              | 4.55                   | 0.35  | 3.03                 | 0.17  |
|                  | <i>a</i> <sub>1</sub> | −2.12                  | 0.30  | −1.80                | 0.24  |
|                  | <i>a</i> <sub>2</sub> | −0.98                  | 0.22  | −1.23                | 0.21  |
|                  | <i>a</i> <sub>3</sub> | −0.59                  | 0.19  | −0.82                | 0.21  |
|                  | <i>b</i>              | 4.52                   | 0.40  | 0.021                | 0.002 |
|                  | <i>c</i>              | 0.981                  | 0.004 | –                    | –     |

The response variables were regressed against the initial (time of planting) distance to the nearest *Kalmia* stem (>0.1 m in height) using nonlinear quantile regression (90th percentile, Eq. 1) for *Pinus* and linear quantile regression (90th percentile, Eq. 2) for *Picea* (all  $p < 0.001$ )

### Black spruce

Black spruce dimensions and growth were adequately modelled using linear quantile regressions (Table 1; Figs. 4, 5), since asymptotic values could not be identified (parameter *b* of Eq. 1 not significant). After 8 growing seasons, the black spruce seedlings that were planted in control plots in close proximity to *Kalmia* stems had a 90th percentile of 72 cm in height, 13 mm in GLD, 5 cm year<sup>−1</sup> in AGR<sub>H</sub> and 1 mm year<sup>−1</sup> in AGR<sub>D</sub>. Fertilization alone increased these values by 13, 41, 32 and 45 % for height, GLD, AGR<sub>H</sub> and AGR<sub>D</sub>, respectively. The 90th percentile of black spruce seedlings planted near *Kalmia* stems in scarified but unfertilized plots was 43 % higher for height, 65 % higher for GLD, 90 % higher for AGR<sub>H</sub> and 77 % higher for AGR<sub>D</sub>, compared to control conditions.



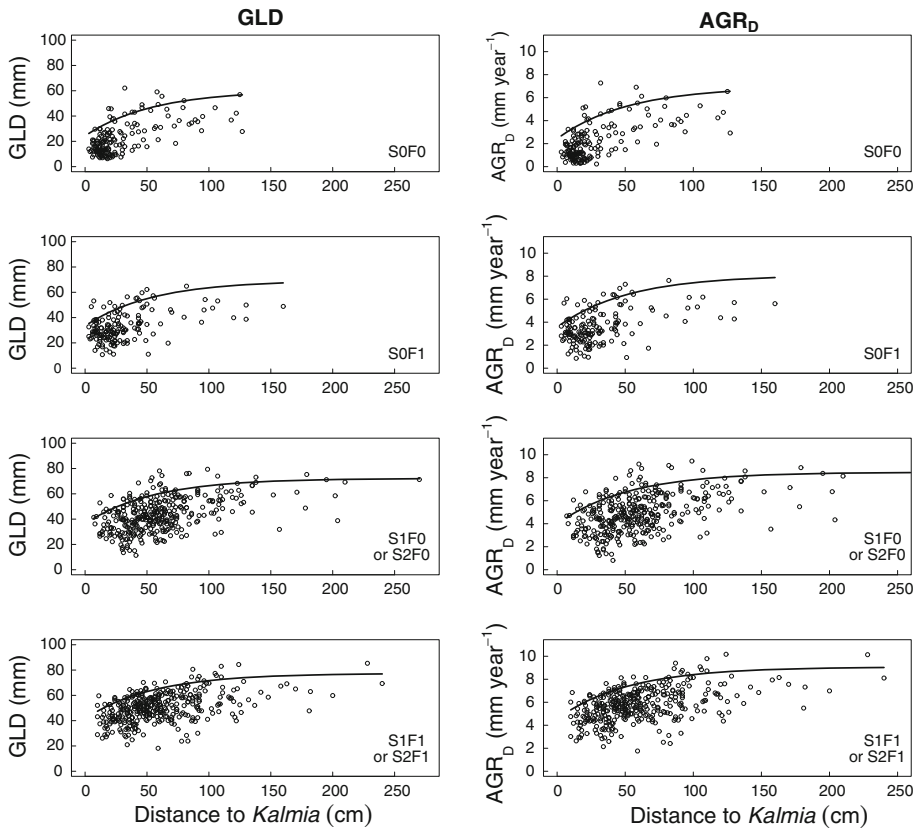
**Fig. 1** Observed (dots) and maximum predicted (lines) 8th-year height and absolute growth rate in height ( $AGR_H$ ) of planted *Pinus banksiana* seedlings, as a function of the initial (time of planting) distance to the nearest *Kalmia* stem ( $>0.1$  m in height) on a clearcut site in northwestern Quebec (Canada). The predicted curves of the 90th percentile are estimated using nonlinear quantile regressions. See Table 1 for estimates of model parameters. S0 = no scarification; S1 = single-pass scarification; S2 = double-pass scarification; F0 = no fertilization; F1 = fertilization at the time of planting

Compared to control plots, the intercepts for height, GLD, and absolute growth rates were increased by 96–180 % in plots that were both scarified and fertilized.

Silviculture and time since treatments interacted with distance to the nearest *Kalmia* stem during the corresponding year to influence black spruce  $AGR_{H-annual}$  (significant interactions in Table 2). The  $AGR_{H-annual}$  was linearly related to the distance of the nearest *Kalmia* stem during growing seasons 2 and 3 in non-fertilized plots (with or without scarification), and in growing seasons 4–5 and 6 in fertilized plots (with or without scarification) (Table 3, Fig. 6).

### *Kalmia*

We detected no interaction between scarification and time (years) regarding the evolution of the distance to the nearest *Kalmia* stems (Table 4). Regardless of the sampling year, the distance to the nearest *Kalmia* stem was significantly lower in unscarified (S0) plots,



**Fig. 2** Observed (dots) and maximum predicted (lines) 8th-year ground-level diameter (GLD) and absolute growth rate in diameter ( $AGR_D$ ) of planted *Pinus banksiana* seedlings, as a function of the initial (time of planting) distance to the nearest *Kalmia* stem ( $>0.1$  m in height) on a clearcut site in northwestern Quebec (Canada). The predicted curves of the 90th percentile are estimated using nonlinear quantile regressions. See Table 1 for estimates of model parameters. S0 = no scarification; S1 = single-pass scarification; S2 = double-pass scarification; F0 = no fertilization; F1 = fertilization at the time of planting

whereas single- and double-pass scarification gave comparable results (Fig. 7). After 2 growing seasons, distance to the nearest *Kalmia* stem had significantly decreased from 46 cm to 17 cm. It remained constant after 3 growing seasons, but the decline had continued by the end of the 5th growing season. It reached 11 cm by the end of 2007, a distance that was significantly less than in the previous seasons.

## Discussion

Our models show that an increased initial planting distance from *Kalmia* leads to greater growth and dimension responses in planted seedlings 8 growing seasons later. For both species, our results thus support previous findings regarding the competitive potential of ericaceous species (de Montigny and Weetman 1990; Mallik 2003; Nilsson et al. 1993; Prescott and Sajedi 2008). *Kalmia* exhibits many morphological traits (allocation to

**Table 2** Results of modeling performed on the annual absolute growth rate in height ( $AGR_{H-annual}$ ) for *Pinus banksiana* and *Picea mariana* seedlings planted on a clearcut site in northwestern Quebec (Canada), as a function of silvicultural treatments and distance to *Kalmia* during the corresponding year

| Source of variation (fixed)   | ndf | <i>Pinus banksiana</i> |       |                  | <i>Picea mariana</i> |       |                |
|-------------------------------|-----|------------------------|-------|------------------|----------------------|-------|----------------|
|                               |     | ddf                    | F     | <i>p</i> value   | ddf                  | F     | <i>p</i> value |
| Scarification (S)             | 1   | 194.0                  | 81.40 | <0.001           | 173.2                | 31.91 | <0.001         |
| Fertilization (F)             | 1   | 173.0                  | 66.45 | <0.001           | 199.0                | 17.90 | <0.001         |
| Year (Y)                      | 4   | 367.4                  | 89.46 | <0.001           | 192.2                | 14.31 | <0.001         |
| S × F                         | 1   | 30.3                   | 4.17  | 0.050            | 45.1                 | 13.25 | 0.001          |
| S × Y                         | 4   | 359.1                  | 21.84 | <0.001           | 179.9                | 7.79  | <0.001         |
| F × Y                         | 4   | 350.3                  | 15.82 | <0.001           | 195.8                | 2.37  | 0.054          |
| Distance to <i>Kalmia</i> (D) | 1   | 471.5                  | 20.55 | <0.001           | 574.9                | 30.33 | <0.001         |
| D × S                         | 1   | 1,218.5                | 6.25  | <b>0.013</b>     | 944.0                | 0.60  | 0.437          |
| D × F                         | 1   | 1,206.8                | 28.63 | <b>&lt;0.001</b> | 543.8                | 0.28  | 0.595          |
| D × Y                         | 4   | 380.1                  | 4.48  | <b>0.002</b>     | 232.0                | 3.14  | 0.015          |
| D × S × F                     | 1   | –                      | –     | –                | 955.4                | 3.89  | <b>0.049</b>   |
| D × F × Y                     | 4   | –                      | –     | –                | 233.0                | 3.53  | <b>0.008</b>   |

With respect to the significant interactions, *p* values in bold are for interpretation (Zar 1999). Denominator degrees of freedom (ddf) were calculated using Satterthwaite's formula (Littell et al. 2006)

*ndf* numerator degrees of freedom

roots, nutrient retention, defence investment, drought and low nutrient tolerance) consistent with a high competitive ability on low productivity sites (Goldberg 1996), such as those found in the boreal forest. This species competes directly for nutrients (Thiffault et al. 2004b), possibly water (LeBel et al. unpublished data), and has numerous indirect effects on soil nutrient availability (Bradley et al. 2000; LeBel et al. 2008; Yamasaki et al. 2002).

As expected, growth responses were species specific: (a) we observed a non-linear relationship between jack pine growth in height and diameter and dimension, and *Kalmia* proximity at the time of planting, and (b) black spruce seedlings responded linearly to the increased distance of the nearest *Kalmia*. The lack of an asymptotic response for black spruce indicates that *Kalmia* influence was still significant at relatively great distances, at least within the radius at which we measured the closest competitor. This result contrasts with those of Yamasaki (1999), who successfully identified a maximum range for *Kalmia* influence on black spruce leader growth for a number of sites in Newfoundland (Canada). However, Yamasaki (1999) reports on *Kalmia* and spruce data that were collected concomitantly (during the 7th growing season), whereas we regressed dimensions at age 8 and growth over 8 growing seasons against initial *Kalmia* presence (year 0) to generate planting recommendations. Yamasaki's results (1999) thus potentially reflect the direct and indirect effects of *Kalmia* on black spruce free growth during a particular year i.e. the elongation of the stem's unit formed during the months when growth occurred (Grossnickle 2000). In contrast, ours integrate the global effect of *Kalmia* over time as a function of its initial position relative to planted seedlings. We indeed detected significant relations between annual growth in height and the corresponding distance to *Kalmia* (Table 2), but the relations varied from year to year, and were greatly influenced by the silvicultural treatments.

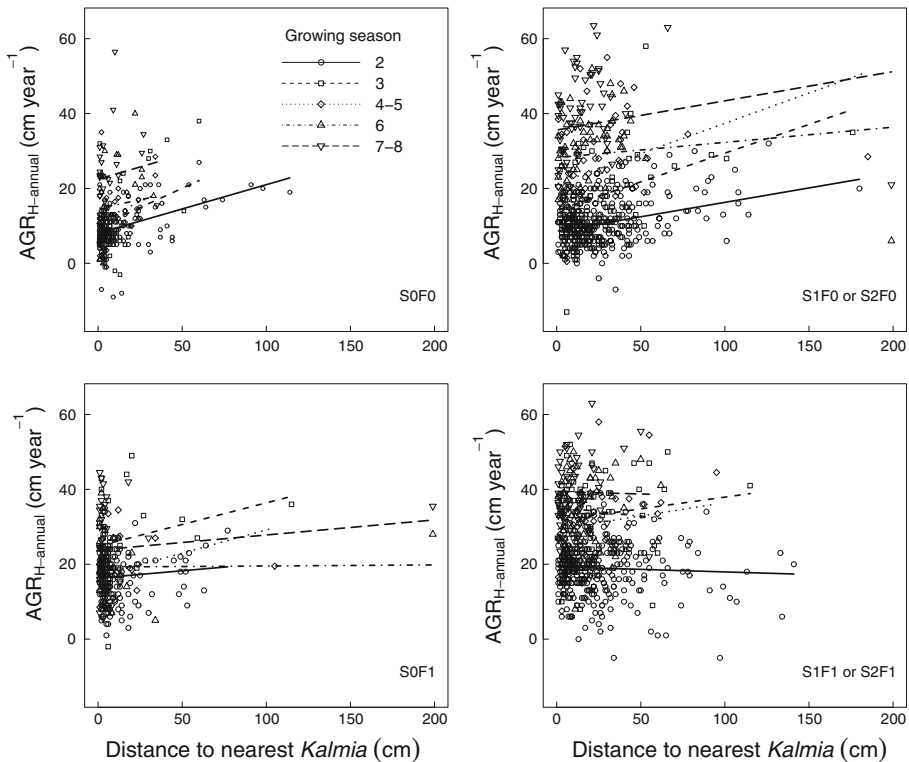
**Table 3** Parameter estimates and standard errors (SE) of the slope related to the distance to nearest *Kalmia* stem (>0.1 m in height) effect on annual absolute growth rate in height (AGR<sub>H-annual</sub>) for *Pinus banksiana* and *Picea mariana* on a clearcut site in northwestern Quebec (Canada)

| Treatment    | Growing season | <i>Pinus banksiana</i> |      |                         | <i>Picea mariana</i> |      |                         |
|--------------|----------------|------------------------|------|-------------------------|----------------------|------|-------------------------|
|              |                | Estimate               | SE   | Adjusted <i>p</i> value | Estimate             | SE   | Adjusted <i>p</i> value |
| S0F0         | 2              | 0.13                   | 0.02 | <b>&lt;0.001</b>        | 0.04                 | 0.01 | <b>0.044</b>            |
|              | 3              | 0.21                   | 0.03 | <b>&lt;0.001</b>        | 0.10                 | 0.03 | <b>0.006</b>            |
|              | 4–5            | 0.21                   | 0.04 | <b>&lt;0.001</b>        | 0.12                 | 0.05 | 0.156                   |
|              | 6              | 0.09                   | 0.03 | 0.051                   | 0.08                 | 0.04 | 0.686                   |
|              | 7–8            | 0.13                   | 0.04 | <b>0.012</b>            | 0.10                 | 0.04 | 0.180                   |
| S0F1         | 2              | 0.04                   | 0.02 | 0.451                   | 0.05                 | 0.02 | 0.190                   |
|              | 3              | 0.12                   | 0.03 | <b>0.006</b>            | 0.07                 | 0.03 | 0.463                   |
|              | 4–5            | 0.12                   | 0.04 | <b>0.014</b>            | 0.16                 | 0.04 | <b>0.001</b>            |
|              | 6              | 0.00                   | 0.03 | 1.000                   | 0.16                 | 0.03 | <b>&lt;0.001</b>        |
|              | 7–8            | 0.04                   | 0.04 | 0.938                   | 0.04                 | 0.03 | 0.728                   |
| S1F0 or S2F0 | 2              | 0.08                   | 0.01 | <b>&lt;0.001</b>        | 0.05                 | 0.01 | <b>&lt;0.001</b>        |
|              | 3              | 0.15                   | 0.03 | <b>&lt;0.001</b>        | 0.12                 | 0.03 | <b>&lt;0.001</b>        |
|              | 4–5            | 0.16                   | 0.04 | <b>&lt;0.001</b>        | 0.14                 | 0.05 | 0.060                   |
|              | 6              | 0.04                   | 0.03 | 0.828                   | 0.09                 | 0.04 | 0.383                   |
|              | 7–8            | 0.08                   | 0.04 | 0.339                   | 0.11                 | 0.04 | 0.056                   |
| S1F1 or S2F1 | 2              | −0.01                  | 0.01 | 0.943                   | 0.01                 | 0.01 | 0.948                   |
|              | 3              | 0.06                   | 0.03 | 0.300                   | 0.03                 | 0.03 | 0.968                   |
|              | 4–5            | 0.07                   | 0.04 | 0.356                   | 0.13                 | 0.04 | <b>0.005</b>            |
|              | 6              | −0.05                  | 0.03 | 0.628                   | 0.12                 | 0.03 | <b>0.001</b>            |
|              | 7–8            | −0.01                  | 0.04 | 1.000                   | 0.01                 | 0.02 | 1.000                   |

Adjusted *p* values (simulation-based approach) in bold are for interpretation (Zar 1999)

0 = No scarification; S1 = single-pass scarification; S2 = double-pass scarification; F0 = no fertilization; F1 = fertilization at the time of planting

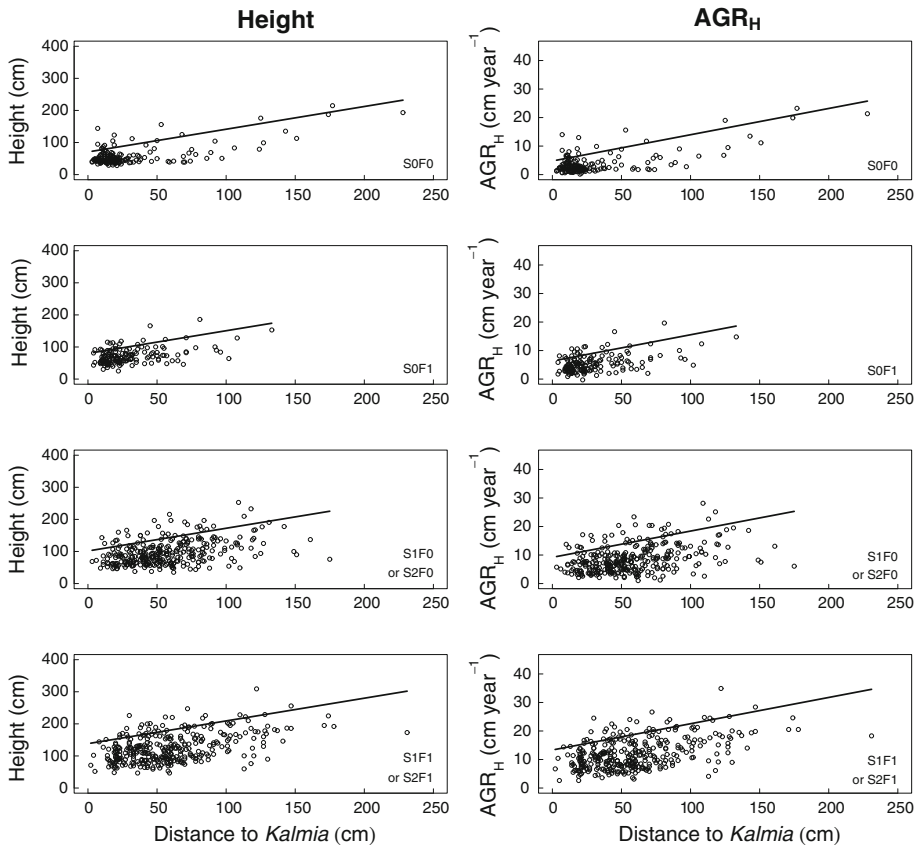
Relative to spruce, pine has a higher potential rate of N uptake by unit root length (Reich et al. 1998), and exhibit an higher N retranslocation efficiency than spruce (Bothwell et al. 2001). We thus hypothesise that this enhanced capacity of pine to absorb and use nutrients limits the extent of *Kalmia* influence on this species relative to black spruce. Moreover, jack pine rooting habits differ greatly from those of black spruce. We believe that this trait might enable jack pine to extend its root into an environment that is relatively free of the influence of *Kalmia*. Whereas jack pine has a strong tap root with most of its root system located at 15–25 cm below the mineral soil surface, black spruce rarely explores the soil below 10 cm (Bannan 1940) and a majority of its roots spread laterally at the moss-humus interface (Viereck and Johnston 1990). Black spruce also shows adventitious root formation in the organic soil (Krause and Morin 2005), a zone heavily colonized by the vegetative propagules of *Kalmia* (Mallik 1993; Thiffault et al. 2004b), and which presumably contains high concentration of phenolic compounds known to reduced primary root growth (Zhu and Mallik 1994). Yamasaki (1999) has indeed evidenced the relationship between humus depth and the intensity of the *Kalmia* effect on planted black spruce. For planted jack pine in scarified plots, Plourde et al. (2009) reported that half of the total root length is unidirectional and found in the scarified furrows, a



**Fig. 3** Observed (symbols) and predicted (lines) annual absolute growth rate in height ( $AGR_{H-annual}$ ) of planted *Pinus banksiana* seedlings, as a function of the distance to the nearest *Kalmia* stem ( $>0.1$  m in height) during the corresponding season, on a clearcut site in northwestern Quebec (Canada). The predicted curves are estimated using the model from Table 2; slopes and their standard errors are provided in Table 3. S0 = no scarification; S1 = single-pass scarification; S2 = double-pass scarification; F0 = no fertilization; F1 = fertilization at the time of planting

microsite that corresponds to one-third of the area surrounding the stems, and in which *Kalmia* regeneration is known to be limited because of the exposed mineral soil (Thiffault et al. 2005; Titus et al. 1995). In comparison, Thiffault (2010) did not observe any impact of trench scarification on the radial distribution of black spruce seedling roots.

Mechanical site preparation can significantly increase soil temperature in the rooting zone, soil water availability, and mineralization rates, especially in boreal forests (Örlander et al. 1990; Prévost 1992). Although the fertilization effect is confined to the first few years following treatment application, and does not influence soil processes as do *Kalmia* eradication (LeBel et al. 2008), it has been shown to positively influence conifer establishment on other ericaceous-dominated sites through its effect on nutrient availability (Prescott and Sajedi 2008; Taylor and Tabbush 1990). Our protocol did not allow separation of these treatment effects from the competition impact of *Kalmia*. However, even with *Kalmia* plants in close range, fertilization and scarification treatments had a beneficial influence on black spruce and jack pine seedling growth and dimensions over 8 growing seasons. Table 3 and Figs. 3 and 6 illustrate how scarification and fertilization affect the distance-dependant relationship between conifer annual growth and *Kalmia*; the slope generally decreases when treated plots are compared to control conditions. Regardless of

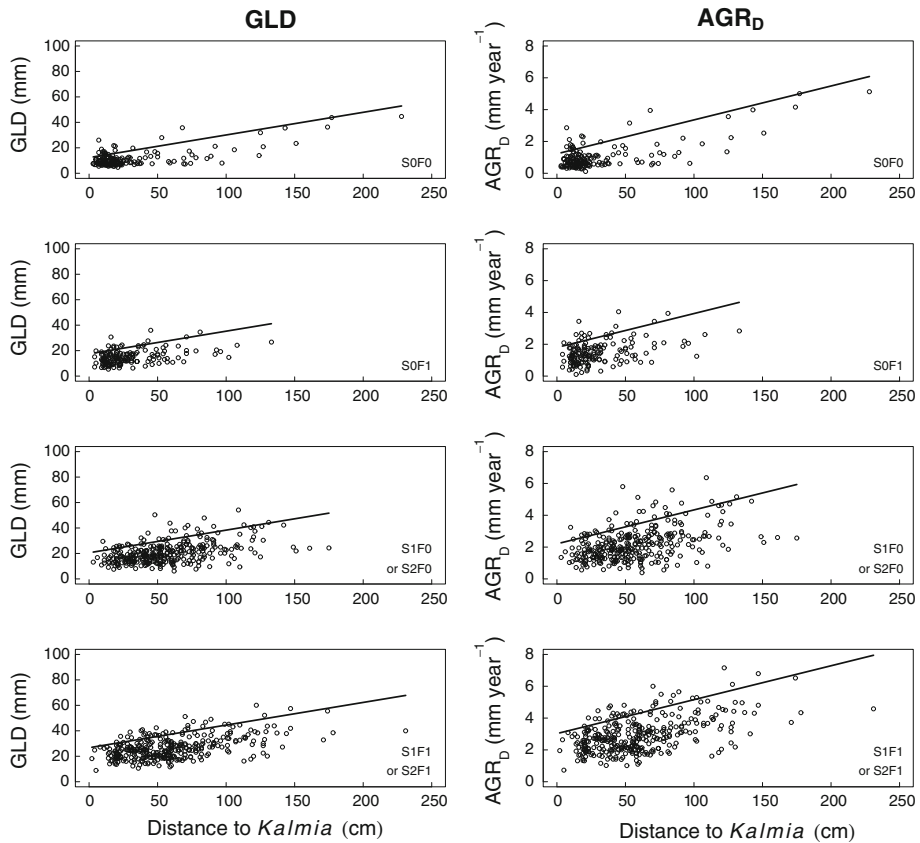


**Fig. 4** Observed (dots) and maximum predicted (lines) 8th-year height and absolute growth rate in height ( $AGR_H$ ) of planted *Picea mariana* seedlings, as a function of the initial (time of planting) distance to the nearest *Kalmia* stem ( $>0.1$  m in height) on a clearcut site in northwestern Quebec (Canada). The predicted curves of the 90th percentile are estimated using linear quantile regressions. See Table 1 for estimates of model parameters. S0 = no scarification; S1 = single-pass scarification; S2 = double-pass scarification; F0 = no fertilization; F1 = fertilization at the time of planting

the distance to *Kalmia*, the scarification and fertilization effect was observed earlier on this same site 3 years following planting (Thiffault et al. 2005). Similar conclusions were drawn from longer-term experiments conducted on other ericaceous-dominated ecosystems in northeastern Canada (Thiffault et al. 2004a; Thiffault et al. 2010).

### Management implications

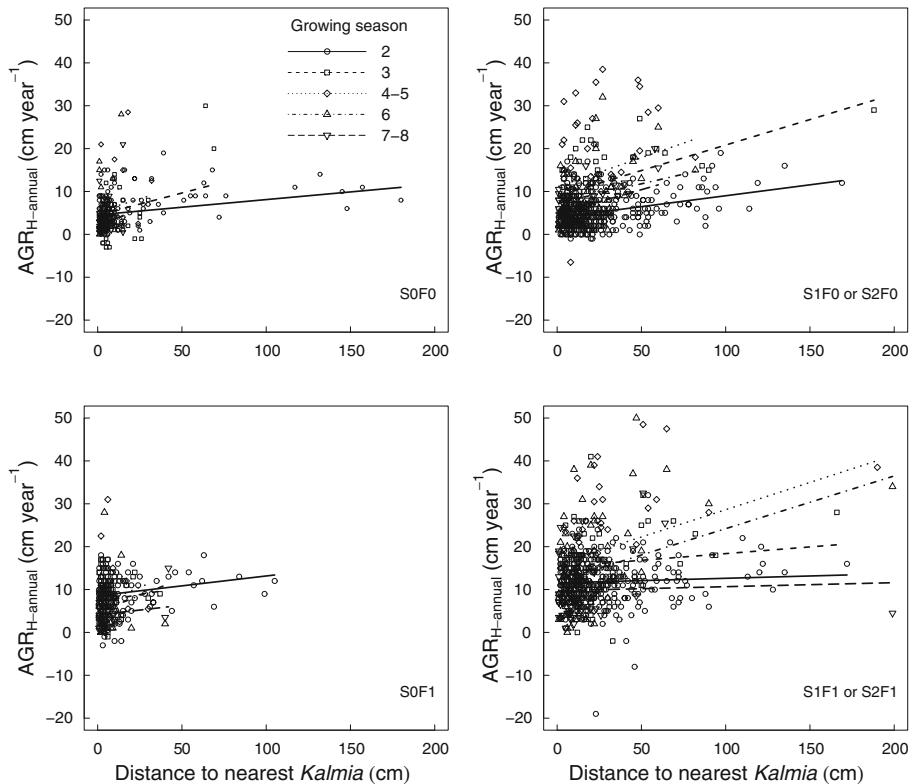
Our main objective was to test if the initial distance to the nearest *Kalmia* stem (an easily-assessed criterion) could be used to identify preferred planting locations on reforestation sites. We acknowledge that other microsite characteristics and competitive interactions are likely to have a significant impact on planted seedling growth during the establishment phase, and that these characteristics and interactions can be expected to vary from site to site (Burney et al. 2007; Delong et al. 1997; Jonsson 1999; Peltzer and Wilson 2001). However, we have concentrated our sampling and analytical efforts on *Kalmia* proximity



**Fig. 5** Observed (dots) and maximum predicted (lines) 8th-year ground-level diameter (GLD) and absolute growth rate in diameter ( $AGR_D$ ) of planted *Picea mariana* seedlings, as a function of the initial (time of planting) distance to the nearest *Kalmia* stem ( $>0.1$  m in height) on a clearcut site in northwestern Quebec (Canada). The predicted curves of the 90th percentile are estimated using linear quantile regressions. See Table 1 for estimates of model parameters. S0 = no scarification; S1 = single-pass scarification; S2 = double-pass scarification; F0 = no fertilization; F1 = fertilization at the time of planting

because other microtopographic variables are already included in planting spot selection in Quebec (MRNF 2006), and *Kalmia* is one of the few significant competitors found in plantations established in the boreal forest of northeastern Canada (Thiffault and Roy 2011). Moreover, our study was conducted on an ecotype typical of reforestation conditions in boreal Quebec. The use of quantile regressions enabled us to model the maximum responses of the seedlings (highest 10th percentile) as a function of initial distance to *Kalmia*. Considering that the social position of trees is likely to be maintained and even reinforced during plantation growth (Jobidon and Charette 1997), our models represent the expected height, GLD and growth of the seedlings that are most likely to become dominant or codominant.

Our results confirmed the nonlinear relationship between jack pine and *Kalmia* proximity. However, black spruce responded linearly to the increased distance of the nearest competitor, which confirms the higher sensitivity of this species to ericaceous vegetation compared to pine. Our results imply that proximity to *Kalmia* can have a significant impact



**Fig. 6** Observed (*symbols*) and predicted (*lines*) annual absolute growth rate in height ( $AGR_{H-annual}$ ) of planted *Picea mariana* seedlings, as a function of the distance to the nearest *Kalmia* stem ( $>0.1$  m in height) during the corresponding season, on a clearcut site in northwestern Quebec (Canada). The predicted *curves* are estimated using the model from Table 2; *slopes* and their standard errors are provided in Table 3. S0 = no scarification; S1 = single-pass scarification; S2 = double-pass scarification; F0 = no fertilization; F1 = fertilization at the time of planting

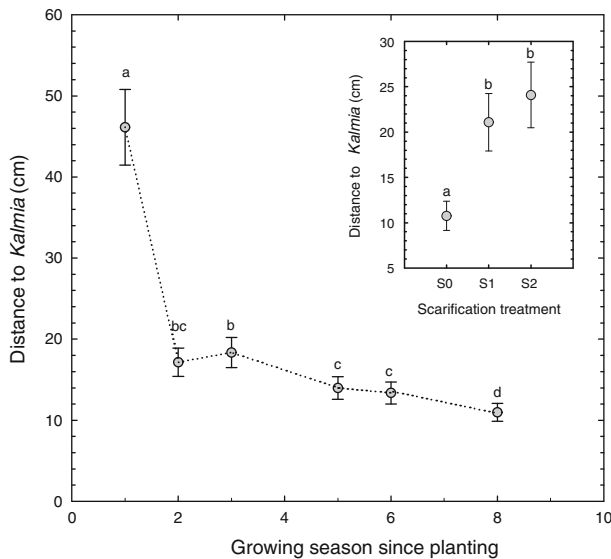
**Table 4** Results of ANOVA performed on the mean distance from planted seedlings to the nearest *Kalmia* stem ( $>0.1$  m in height) on a clearcut site in northwestern Quebec (Canada)

| Source of variation (fixed) | Initial distance to the nearest <i>Kalmia</i> stem |       |       |                  |
|-----------------------------|----------------------------------------------------|-------|-------|------------------|
|                             | ndf                                                | ddf   | F     | <i>p</i> value   |
| Scarification               | 2                                                  | 30.1  | 8.34  | <b>0.001</b>     |
| Year                        | 5                                                  | 129.1 | 93.81 | <b>&lt;0.001</b> |
| Scarification $\times$ Year | 10                                                 | 129.1 | 1.21  | 0.292            |

With respect to the significant interactions, *p* values in bold are for interpretation (Zar 1999)

Denominator degrees of freedom (ddf) were calculated using Satterthwaite's formula (Littell et al. 2006)  
*ndf* numerator degrees of freedom

on future seedling dimensions, most probably by reducing the competitive interactions during the first few years following planting (Thiffault et al. 2010). Hence planting microsites should be chosen accordingly.



**Fig. 7** Distance between the nearest *Kalmia* stem (>0.1 m in height) and the planted seedlings as influenced by time (*main frame*) and scarification (*insert*) on a clearcut site in northwestern Quebec (Canada). See Table 2 for *p* values. Values identified by the *same letters* are not significantly different at  $\alpha = 0.05$ . S0 = no scarification; S1 = single-pass scarification; S2 = double-pass scarification

For jack pine, we can use the model parameters to predict the 90th percentile seedling height and growth (i.e. the asymptote) for each treatment combination and calculate the percentage of maximum height and growth that will be attained at given initial distances from *Kalmia*. Table 5 can be used to estimate the loss in height or growth associated with a given initial distance from *Kalmia* at the time of planting. As an example, when planting jack pine on scarified (non-fertilized) sites, a *Kalmia*-free radius of 60 cm would lead to ~80 to 85 % (between 277 and 294 cm) of the 90th percentile of maximum potential height at age 8 years for this treatment (Table 5). When planting on non-scarified (non-fertilized) sites, the same *Kalmia*-free criteria would lead to ~75 to 80 % (between 219 and 234 cm) of the 90th percentile of maximum potential height at age 8 years that can be expected in the absence of treatment. The use of a fertilization treatment at the time of planting can reduce by 5–11 cm the *Kalmia*-free radius required to reach a given percent of full potential height.

Height is preferred to diameter for these calculations, as the model parameters (Table 1) confirm that apical growth is more sensitive to *Kalmia* proximity than radial growth. In other words, distance criteria would be reduced if based on diameter rather than on height (e.g. 97 cm to reach 90 % of the full  $AGR_D$  potential in S0F0 plots, as compared to 134 cm to reach 90 % of the full  $AGR_H$ ). Diameter growth is indeed a sensitive indicator of competitive interactions, especially in contexts of competition for light (Jobidon 2000). But, previous studies conducted on the same site (Thiffault et al. 2004b, 2005) have confirmed that both height and diameter growth are affected in these conditions, as competition for nutrients and allelopathic interactions (rather than light competition) are the main drivers of seedling performance in this context.

Finally, our results show that scarification created *Kalmia*-free microsites that were rapidly re-invaded by vegetative propagules in the following years. This result stresses the

**Table 5** Predicted initial (time of planting) distance to the nearest *Kalmia* stem (>0.1 m in height) at which *Pinus banksiana* seedlings reach a given percent of predicted 90th percentile maximum 8th-year height or absolute growth rate in height over 8 years (AGR<sub>H</sub>) on a clearcut site in northwestern Quebec (Canada)

|                                                                              | S0F0          |                  | S0F1   |                  | S1F0 or S2F0 |                  | S1F1 or S2F1 |                  |
|------------------------------------------------------------------------------|---------------|------------------|--------|------------------|--------------|------------------|--------------|------------------|
|                                                                              | Height        | AGR <sub>H</sub> | Height | AGR <sub>H</sub> | Height       | AGR <sub>H</sub> | Height       | AGR <sub>H</sub> |
| Asymptote [maximum height (cm) or AGR <sub>H</sub> (cm year <sup>-1</sup> )] | 292           | 33               | 336    | 39               | 346          | 40               | 375          | 44               |
| Percent of asymptote (%)                                                     | Distance (cm) |                  |        |                  |              |                  |              |                  |
| 65                                                                           | 24            | 34               | 13     | 22               | 11           | 20               | 6            | 12               |
| 70                                                                           | 35            | 46               | 25     | 35               | 22           | 32               | 17           | 25               |
| 75                                                                           | 48            | 61               | 38     | 49               | 35           | 47               | 30           | 39               |
| 80                                                                           | 64            | 79               | 53     | 67               | 51           | 65               | 46           | 57               |
| 85                                                                           | 84            | 102              | 74     | 90               | 72           | 88               | 66           | 80               |
| 90                                                                           | 113           | 134              | 103    | 123              | 101          | 120              | 95           | 113              |

See Table 1 for model parameters

S0 = No scarification; S1 = single-pass scarification; S2 = double-pass scarification; F0 = No fertilization; F1 = Fertilization at the time of planting

importance of rapidly proceed to reforestation activities following site preparation, as available microsites disappear with time and planting in close proximity to *Kalmia* has a significant impact on growth.

**Acknowledgments** We are indebted to Francis Cadoret, Réjean Poliquin, Jacques Carignan, Steve Lemay, Dominic Létourneau, Evelyne Gaillard, Daniel Boudreault (deceased), Philippe LeBel, and several summer students who contributed to the establishment and measurement of the experimental plots over time. We thank Dr. BD Titus for providing helpful comments on an earlier version of the manuscript and Dr. SC Grossnickle for insightful discussions regarding boundary line analyses. Two anonymous reviewers, the Guest Editor and the Editor-in-Chief also provided constructive advices on an earlier draft. We acknowledge the contribution of Produits Forestiers Résolu (then Abitibi Consolidated) and the regional staff of the *Ministère des Ressources naturelles et de la Faune du Québec* (MRNFQ) during the establishment phase of the project. This study is part of project 112310031 of the *Direction de la recherche forestière*, MRNFQ.

## References

- Balandier P, Collet C, Miller JH, Reynolds PE, Zedaker SM (2006) Designing forest vegetation management strategies based on the mechanisms and dynamics of crop tree competition by neighbouring vegetation. *Forestry* 79(1):3–27
- Bannan MW (1940) The root systems of northern Ontario conifers growing in sand. *Am J Bot* 27(2): 108–114
- Bothwell KS, Prescott CE, Jones MD (2001) Factors contributing to the superior growth and N nutrition of 11-year-old lodgepole pine compared with Sitka spruce on a N-poor cedar-hemlock cutover. *Can J For Res* 31(7):1272–1279
- Bradley RL, Titus BD, Preston CP (2000) Change to mineral N cycling and microbial communities in black spruce humus after additions of (NH<sub>4</sub>)<sub>2</sub>SO<sub>4</sub> and condensed tannins extracted from *Kalmia angustifolia* and balsam fir. *Soil Biol Biochem* 32(8/9):1227–1240
- Brown H, Prescott R (2006) *Applied mixed models in medicine*, 2nd edn. Wiley, Mississauga, ON
- Burney O, Wing MG, Rose R (2007) Microsite influences on variability in Douglas-fir seedling development. *West J Appl For* 22(3):156–162

- Burton PJ, Bedford L, Goldstein M, Osberg M (2000) Effects of disk trench orientation and planting spot position on the ten-year performance of lodgepole pine. *New For* 20(1):23–44
- Canadian Council of Forest Ministers (2009) National forestry database program. Available online (<http://nfdp.ccfm.org>)
- de Montigny LM, Weetman GF (1990) The effects of ericaceous plants on forest productivity. Canadian forest service, forestry Canada. Information report N-X-271. St. John's, NF
- Delong HB, Lieffers VJ, Blenis PV (1997) Microsite effects on first-year establishment and overwinter survival of white spruce in aspen-dominated boreal mixedwoods. *Can J For Res* 27(9):1452–1457
- Evans J (ed) (2009) Planted forests. Uses, impacts and sustainability. CAB International and FAO, Rome
- Ganatsas P, Tsitsoni T, Tsakalidimi M, Zagas T (2012) Reforestation of degraded Kermes oak shrublands with planted pines: effects on vegetation cover, species diversity and community structure. *New For* 43(1):1–11
- Goldberg DE (1996) Competitive ability: definitions, contingency and correlated traits. *Philos Trans R Soc Lond B Biol Sci* 351(1345):1377–1385
- Grossnickle SC (2000) Ecophysiology of Northern spruce species: the performance of planted seedlings. NRC Research Press, Ottawa, ON
- Grossnickle SC, Arnott JT (1992) Gas exchange response of western hemlock seedlings from various dormancy-induction treatments to reforestation site environmental conditions. *For Ecol Manag* 49(3–4):177–193
- Heineman J (1998) Forest floor planting: a discussion of issues as they relate to various site-limiting factors. Forest site management section, forest practices branch. Silviculture note 16. Victoria, BC
- Jarvis PG (1976) Interpretation of variations in leaf water potential and stomatal conductance found in canopies in field. *Philos Trans R Soc Lond B Biol Sci* 273(927):593–610
- Joannis GD, Bradley RL, Preston CM, Bending GD (2009) Sequestration of soil nitrogen as tannin-protein complexes may improve the competitive ability of sheep laurel (*Kalmia angustifolia*) relative to black spruce (*Picea mariana*). *New Phytol* 181(1):187–198
- Jobidon R (2000) Density-dependent effects of northern hardwood competition on selected environmental resources and young white spruce (*Picea glauca*) plantation growth, mineral nutrition, and stand structural development: a 5-year study. *For Ecol Manag* 130(1–3):77–97
- Jobidon R, Charette L (1997) Effets, après 10 ans, du dégagement manuel simple ou répété et de la période de coupe de la végétation de compétition sur la croissance de l'épinette noire en plantation. *Can J For Res* 27(12):1979–1991
- Jonsson B (1999) Stand establishment and early growth of planted *Pinus sylvestris* and *Picea abies* related to microsite conditions. *Scand J For Res* 14(5):425–439
- Koenker R (2005) Quantile Regression. Cambridge University Press, New York, NY
- Koenker R (2009) Quantile regression in R: a vignette. Available online (<http://www.econ.uiuc.edu/~roger/research/rq/vig.pdf>)
- Koenker R, Hallock KF (2001) Quantile regression. *J Econ Perspect* 15(4):143–156
- Krause C, Morin H (2005) Adventive-root development in mature black spruce and balsam fir in the boreal forests of Quebec, Canada. *Can J For Res* 35(11):2642–2654
- LeBel P, Thiffault N, Bradley RL (2008) *Kalmia* removal increases nutrient supply and growth of black spruce seedlings: an effect fertilizer cannot emulate. *For Ecol Manag* 256(10):1780–1784
- Littell RC, Milliken GA, Stroup WW, Wolfinger RD, Schabenberger O (2006) SAS system for mixed models, 2nd edn. SAS Institute Inc, Cary, NC
- Mallik AU (1993) Ecology of a forest weed of Newfoundland: vegetative regeneration strategy of *Kalmia angustifolia*. *Can J Bot* 71(1):161–166
- Mallik AU (2003) Conifer regeneration problems in boreal and temperate forests with ericaceous understory: role of disturbance, seedbed limitation, and keystone species change. *Crit Rev Plant Sci* 22(3–4):341–366
- Margolis HA, Brand DG (1990) An ecophysiological basis for understanding plantation establishment. *Can J For Res* 20(4):375–390
- Moroni MT, Thiffault N, Titus BD, Mante C, Makeschin F (2009) Controlling *Kalmia* and reestablishing conifer dominance enhances soil fertility indicators in central Newfoundland and Canada. *Can J For Res* 39(7):1270–1279
- MRNF (2006) Qualité des plantations—guide de l'évaluateur. Ministère des ressources naturelles et de la Faune, direction de l'assistance technique et direction des traitements sylvicoles. Code de diffusion 2005-3015. Québec, QC
- Newton M (2006) Taking charge in forest vegetation management. *Can J For Res* 36(10):2357–2363

- Nilsson MC, Högberg P, Zackrisson O, Fengyou W (1993) Allelopathic effects by *Empetrum hermaphroditum* on development and nitrogen uptake by roots and mycorrhizae of *Pinus silvestris*. *Can J Bot* 71(4):620–628
- Örlander G, Gemmel P, Hunt J (1990) Site preparation: a Swedish overview. British Columbia Ministry of Forests. FRDA Report 105. Victoria, BC
- Paquette A, Messier C (2010) The role of plantations in managing the world's forests in the Anthropocene. *Front Ecol Environ* 8(1):27–34
- Peltzer DA, Wilson SD (2001) Variation in plant responses to neighbors at local and regional scales. *Am Nat* 157(6):610–625
- Pienarr LV, Turnbull KJ (1973) The Chapman-Richards generalized of Von Bertalanffy's growth model for basal area growth and yield in even-aged stands. *For Sci* 19(1):2–22
- Plourde A, Krause C, Lord D (2009) Spatial distribution, architecture, and development of the root system of *Pinus banksiana* Lamb. in natural and planted stands. *For Ecol Manag* 258(9):2143–2152
- Prégent G, Végiard S (2000) Rendement anticipé des plantations d'épinette noire dans les domaines écologiques de la pessière noire. Ministère des Ressources naturelles. Note de recherche forestière 109. Québec, QC
- Prescott CE, Sajedi T (2008) The role of salal in forest regeneration problems in coastal British Columbia: problem or symptom? *For Chron* 84(1):29–36
- Prévost M (1992) Effets du scarifiage sur les propriétés du sol, la croissance des semis et la compétition: revue des connaissances actuelles et perspectives de recherches au Québec. *Ann Sci For* 49(3):277–296
- R Core Team (2011) The R project for statistical computing. Revised on October 25, 2011. Available online (<http://www.r-project.org/>)
- Reich PB, Walters MB, Tjoelker MG, Vanderklein D, Buschena C (1998) Photosynthesis and respiration rates depend on leaf and root morphology and nitrogen concentration in nine boreal tree species differing in relative growth rate. *Funct Ecol* 12(3):395–405
- Richards FJ (1959) A flexible growth function for empirical use. *J Exp Bot* 10(29):290–300
- Robitaille A, Saucier JP (1998) Paysages régionaux du Québec méridional. Gouvernement du Québec, Ministère des Ressources naturelles, Québec, QC
- SAS Institute Inc (2007) Statistical analysis with the GLIMMIX procedure. SAS Institute Inc, Cary, NC
- SAS Institute Inc (2010) SAS/STAT® 9.2 user's guide, 2nd edn. SAS Institute Inc, Cary, NC
- Saucier JP, Robitaille A, Grondin P (2009) Cadre bioclimatique du Québec. In *Écologie forestière. Manuel de foresterie*, 2nd edn. Ordre des ingénieurs forestiers du Québec, Éditions Multimondes, Québec, QC, pp 186–205.
- Schumacher FX (1939) A new growth curve and its application to timber yield studies. *J For* 37:819–820
- Soil Classification Working Group (1998) The Canadian system of soil classification. 3rd edn. Agriculture and Agri-Food Canada. Publication 1646. Ottawa, ON
- Sutherland BJ, Foreman FF (1995) Guide to the use of mechanical site preparation equipment in North-western Ontario. Canadian Forest Service, natural resources Canada. Technical report TR-87. Sault Ste. Marie, ON
- Taylor CMA, Tabbush PM (1990) Nitrogen deficiency in Sitka spruce plantations. *For Comm (UK). Bull.* 89
- Thiffault N (2010) Stabilité mécanique et caractéristiques racinaires de plants de fortes dimensions de *Picea mariana* produits en récipients ou à racines nues. *For Chron* 86(4):469–476
- Thiffault N, Roy V (2011) Living without herbicides in Québec (Canada): historical context, current strategy, research and challenges in forest vegetation management. *Eur J For Res* 130(1):117–133
- Thiffault N, Cyr G, Prégent G, Jobidon R, Charette L (2004a) Artificial regeneration of an ericacees black spruce stand—effects of scarification, fertilization and plant type 10 years later. *For Chron* 80(1):141–149
- Thiffault N, Titus BD, Munson AD (2004b) Black spruce seedlings in a *Kalmia-Vaccinium* association: microsite manipulation to explore interactions in the field. *Can J For Res* 34(8):1657–1668
- Thiffault N, Titus BD, Munson AD (2005) Silvicultural options to promote seedling establishment on *Kalmia-Vaccinium*-dominated sites. *Scand J For Res* 20(2):110–121
- Thiffault N, Titus BD, Moroni MT (2010) Silviculture and planted species interact to influence reforestation success on a *Kalmia*-dominated site—a 15-year study. *For Chron* 86(2):234–242
- Thompson DG, Pitt DG (2003) A review of Canadian forest vegetation management research and practice. *Ann For Sci* 60(7):559–572
- Titus BD, Sidhu SS, Mallik AU (1995) A summary of some studies on *Kalmia angustifolia* L.: a problem species in Newfoundland forestry. Canadian Forest Service. Information report N-X-296. St. John's, NF
- Viereck LA, Johnston WF (1990) *Picea mariana* (Mill.) B. S. P. in Silvics of North America. In: Burns RM, Honkala BH (eds) *Conifers*, vol. 1. USDA Forest Service, Agriculture Handbook 654, Washington, DC

- Yamasaki SH (1999) Influences of *Kalmia angustifolia* on black spruce in eastern Canada's boreal forest. Ph.D. Thesis, McGill University, Montréal, QC
- Yamasaki SH, Fyles JW, Egger KN, Titus BD (1998) The effect of *Kalmia angustifolia* on the growth, nutrition, and ectomycorrhizal symbiont community of black spruce. *For Ecol Manag* 105(1–3): 197–207
- Yamasaki SH, Fyles JW, Titus BD (2002) Interactions among *Kalmia angustifolia*, soil characteristics, and the growth and nutrition of black spruce seedlings in two boreal Newfoundland plantations of contrasting fertility. *Can J For Res* 32(12):2215–2224
- Yildiz O, Esen D, Zedaker SM (2010) Five-year effects of cutting and herbicide treatments on control of *Rhododendron flavum* Don., and macronutrient pools in eastern beech (*Fagus orientalis* Lipsky) forests of Turkey. *New For* 40(2):175–184
- Zar JH (1999) Biostatistical analysis, 4th edn. Prentice Hall, Upper Saddle River, NJ
- Zhu H, Mallik AU (1994) Interactions between *Kalmia* and black spruce: isolation and identification of allelopathic compounds. *J Chem Ecol* 20(2):407–421